Draft version November 18, 2019
Typeset using LaTeX **twocolumn** style in AASTeX62

Quenching timescales of dwarf satellites around Milky Way-mass hosts

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ABSTRACT

We present measurements of the quenched fractions and quenching timescales of dwarf satellite galaxies, derived from the Justice League suite of 4 ultra-high-resolution cosmological simulations of Milky Way-mass halos. We show that these simulations accurately reproduce the satellite luminosity functions of observed nearby galaxies, as well as variations in quenched fractions across stellar mass. We trace the histories of satellite galaxies back to $z \sim 15$, and find that most intermediate-mass satellites quench within 3 Gyr of infall into the host halo. A characteristic stellar mass scale of 10^8 $M_\odot$ is apparent, above which infalling satellites are largely resistant to rapid environmental quenching.

Keywords: galaxies: evolution – galaxies: star formation – galaxies: dwarf – galaxies: luminosity function – cosmology: reionization

1. INTRODUCTION

The small potentials of dwarf galaxies make them uniquely sensitive laboratories for understanding the physics of galaxy formation. In particular, their sensitivity to their environment makes dwarfs useful in studying the quenching of star-formation. Observations show that nearly all Local Group (LG) satellites with $M_* < 10^8$ $M_\odot$ are quenched (Phillips et al. 2014; Wetzel et al. 2015; Weisz et al. 2015), but almost no isolated dwarfs are (Geha et al. 2012). This dichotomy is supported further by observations of the gas content of LG dwarfs, with atomic hydrogen (H I) undetected in all satellites within $\sim$ 270 kpc of their host (Grcevich & Putman 2009; Spekkens et al. 2014). While quenching may act differently outside of the Local Group (see Geha et al. 2017), it is well established that the the halo environment of a massive LG galaxy can greatly affect the star-formation histories of its dwarf satellites.

The large number of quenched satellites in the Local Group implies the role of an environmental quenching process that acts rapidly upon infall into the host halo. Traditionally, rapid removal of cold disk gas by ram pressure and tidal stripping is seen as the dominant quenching mechanism, and has been shown to efficiently quench infalling dwarfs in many simulations (e.g. Mayer et al. 2006; Slater & Bell 2014; Bahé & McCarthy 2015; Kazantzidis et al. 2017; Simpson et al. 2018). Stellar feedback may contribute to this process, both by increasing the efficiency of stripping by expelling and heating central gas (Bahé & McCarthy 2015; Kazantzidis et al. 2017) and by reducing the central satellite mass and therefore reducing its potential (Zolotov et al. 2012). Additionally, at higher masses, strangulation—or the suppressed gas accretion due to the precense of a massive halo—may be able to efficiently quench satellites (McGee et al. 2014; Wheeler et al. 2014). While the effects of these individual processes are well-documented, it remains unclear how they conspire to quench dwarf satellites across mass scales.

In order to disentangle the roles of quenching processes, many studies have estimated the quenching timescales necessary to reproduce the quenched fractions observed in the Local Group. For low-mass satellites ($M_* < 10^8$ $M_\odot$), quenching must occur rapidly (1–2 Gyr) after infall (Slater & Bell 2014; Fillingham et al. 2015; Wetzel et al. 2015), a timescale largely consistent with ram pressure stripping (for a contrasting view, see Emerick et al. 2016). For more massive satellites, in which ram pressure stripping is likely inefficient, estimates show much longer quenching timescales ($\gtrsim$ 8 Gyr), more consistent with strangulation models, in which quenching occurs on the cold gas depletion timescale (Wheeler et al. 2014). The variation of the quenching timescale across mass scales can tell us about the quenching efficiency and thus the potential processes at play.

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Table 1. Host Halo Properties

Sim	$M_{\rm vir}$ ($M_\odot$)	M_* ($M_\odot$)	$M_{\rm gas}$ ($M_\odot$)	$R_{\rm vir}$ (kpc)
Sandra	2.7×10^{12}	1.9×10^{11}	2.0×10^{11}	432
Ruth	1.3×10^{12}	1.0×10^{11}	9.3×10^{10}	340
Sonia	1.2×10^{12}	9.0×10^{10}	9.7×10^{10}	333
Elena	8.3×10^{11}	9.0×10^{10}	3.5×10^{10}	293

At present, most studies of the quenching timescale are observational in nature, and use simulations and abundance-matching techniques only to constrain the infall times of Local Group dwarfs (e.g. Wetzel et al. 2015; Fillingham et al. 2015). While this method leads to large uncertainties in the infall time, deriving the infall and quenching times directly from simulations minimizes this uncertainty and avoids assuming that the Local Group is indicative of quenching in general. There are, of course, studies using simulations to directly compute the quenching timescale. However, many of these simulations achieve high resolution only by focusing on individual satellites (e.g. Mayer et al. 2006; Emerick et al. 2016), and others that simulate larger cosmological volumes do so at the cost of resolution (e.g. Bahé & McCarthy 2015; Simpson et al. 2018). Simulating individual satellites inhibits direct comparison to observation, and high-resolution is important to understand quenching across the full range of dwarf satellite masses. In this paper, we use the DC Justice League of simulations to balance these demands by simulating large volumes at ultra-high resolution, such that all dwarfs surviving reionization are resovled.

Specifically, we analyze the mass-dependence of the quenching timescale in dwarf satellites around four Milky-Way analog hosts. The structure of the paper is as follows. In §2, we present the suite of simulations. In §3 we validate the simulations with comparisons to observations. In §4 we present the quenching and infall times of our dwarf satellites. In §5 we discuss the implications of our results for future study, and we present our conclusions in §6.

2. METHODS

We conducted our analysis on the DC Justice League suite of 4 ultra-high resolution, cosmological, smooth particle hydrodynamics (SPH) simulations of Milky Way-mass halos and their surroundings. The properties of the host halos in the Justice League simulations are shown in Table 1. Here we discuss the simulation code and the post-processing analysis conducted on the simulation output.

2.1. *Simulations*

All simulations are generated using the tree+SPH code CHANGA (Menon et al. 2015), which scales efficiently up to 100,000 cores. CHANGA is the successor to the N-body gravity-tree code, PKDGRAV (Stadel 2001) and SPH code GASOLINE. CHANGA models Kelvin Helmholtz instabilities in shearing flows by using the geometric mean density in the SPH force expression (Ritchie & Thomas 2001). It also allows for thermal diffusion across gas particles (Shen et al. 2010).

The simulations were integrated to redshift 0 in a fully cosmological context assuming Planck 2015 cosmological parameters ($\Omega_0 = 0.3086$, $\Lambda = 0.6914$, $h = 0.67$, $\sigma_8 = 0.77$; Planck Collaboration 2016). In order to achieve ultra-high resolution while maintaining the effects of the large-scale environment, the initial conditions were generated using the "zoom-in" volume renormalization technique (Katz & White 1993). The main halos were selected from 50^3 Mpc^3 dark matter-only volumes based on their XXXX. The resolved region extends to ≈ 1.5 times the redshift 0 virial radius of the main halo. The simulations have a force resolution of 87 pc and gas (star) particle masses of 3300 (994) $M_\odot$, allowing us to resolve halos down to $M_{\rm vir} \sim 10^6$ $M_\odot$.

CHANGA follows non-equilibrium abundances of H (including H_2) and He species. The integration of these chemical species and the associated heating and cooling is described in Christensen et al. (2012) and references therein. Photoionization and heating rates are implemented using a uniform, time-dependent cosmic UV background adapted from Haardt & Madau (2012). In this model, cosmological HII regions overlap at $z \sim 6.7$ (13 Gyr ago), indicating the onset of reionization. This reionization model is known to XXX. Additional cooling is provided by metal lines (Shen et al. 2010). Oxygen and iron are tracked independently, and metals are diffused based on a subgrid turbulent mixing model (Shen et al. 2010) with a metal diffusion constant of 0.03.

Star formation in CHANGA is implemented probabilistically according to local H_2 abundance, gas density, and gas temperature, as described in Christensen et al. (2012). Briefly, the star formation efficiency factor, $c^* = 0.1 X_{\rm H_2} = \frac{X_{\rm H_2}}{X_{\rm H_2} + X_{\rm HI}}$, where $X_{\rm HI}$ and $X_{\rm H_2}$ are the mass fraction of atomic and molecular hydrogen, respectively. Star formation is only allowed for particles with temperatures $< 10^3$ K and densities > 0.1 amu cm^{-3}, although these constraints are largely superseded by the dependency of c^* on H_2.

Energy from Type II supernovae (SNe II) is distributed as thermal energy to surrounding gas particles according to the "blastwave" model (Stinson et al. 2006), assuming the canonical $\eta_{\rm SN} = 1.5 \times 10^{51}$ ergs released per SN. Cooling of the affected gas particles is

temporarily disabled to match the theoretical timescale of the "snowplow" phase (McKee & Ostriker 1977) and to avoid rapid radiative cooling that would render feedback negligible. This supernovae feedback model provides the entirety of stellar feedback, as radiative pressure (e.g. Stinson et al. 2012; Hopkins et al. 2014) and other forms of early stellar feedback are not separately included. This stellar feedback model relies entirely on the local properties of the gas, rather than the overall halo potential, but still produces mass-loading factors consistent with energy driven winds (Christensen et al. 2016). It has been critical for reproducing the observed properties of dwarf galaxies, including cored dark matter profiles (Governato et al. 2010), and solving the Too Big To Fail problem (Brooks & Zolotov 2014).

This version of ChaNGa also include supermassive black hole formation (Bellovary et al. 2011), growth, mergers and feedback (Tremmel et al. 2017). However, only X of X dwarf galaxies contain black holes at a redshift of zero, and none accrete at high enough rates during their history for black holes to play a critical role in the quenching of star formation

2.2. *Post-processing Analysis*

In order to select individual halos from simulation snapshots, we used amiga's halo finder (Knollmann & Knebe 2009), which identifies regions of overdensity and assigns halo ownership to gravitationally bound particles. We compute the virial radius ($R_{\rm vir}$) of each halo as the radius at which the enclosed density drops below 200 times the background density, ρ_b. Halo properties are calculated from all particles within the virial radius, excluding subhalos. Table 1 shows basic $z = 0$ properties of the host halos in the Justice League simulations, and Figure 1 shows line-of-sight averaged density plots of the dark matter, gas, and stars in Sandra and Sonia.

For each Justice League simulation, we used the database-generation software tangos (Pontzen & Tremmel 2018) to track particles across snapshots and generate merger trees for all halos containing stars at $z = 0$. For each halo, we determine the major progenitors, defined as the halo which contains the majority of particles from the corresponding halo in the subsequent snapshot, back to $z \approx 15$. We limited the merger tree to progenitor halos with > 1000 dark matter particles, corresponding to a halo mass of $\sim 10^6$ $\mathrm{M}_\odot$, approximately the resolution limit of our simulations. All further analysis was conducted in Python using the pynbody package (Pontzen et al. 2013).

3. OBSERVATIONAL COMPARISONS

The Justice League simulations include advanced physics modules and are some of the highest-resolution ever run; however, their predictive ability is dependent on their accuracy in reproducing observations.

Figure 2 shows the cumulative satellite luminosity functions (LFs) of our four Justice League simulations, alongside observed LFs for nearby host-satellite systems. Although the LFs of our simulations vary widely (from ~ 3 to ~ 15 satellites at $M_V < -10$), they are consistent with the range of observed LFs. The most populous system, Sandra, is consistent with the LFs of massive galaxies such as Centaurus A, M31, and M81. The less massive hosts are most consistent with the MW and M101, and the least massive host, Elena, is consistent with the unique and sparsely populated M94 system.

Our simulations are not susceptible to the missing satellite problem, as we do not see systematic overproduction of luminous satellites. While each galaxy hosts on the order of 10-50 dark matter subhalos with $M_{\rm vir} > 10^8$ $\mathrm{M}_\odot$, the occupation fraction for the least massive halos ($M_{\rm vir} < 10^8$ $\mathrm{M}_\odot$) is low, with $< 5\%$ of halos hosting > 10 star particles. This is a result of the inclusion of baryonic physics in the simulation, which has been shown to reduce the number of satellites (Brooks et al. 2013). Additionally, the LFs of our simulated sample are generally ordered by host halo mass—that is, on the faint end ($M_V > -10$), the more massive halo hosts more luminous satellites. This trend is to be expected (Trentham & Tully 2009; Nickerson et al. 2013). Higher resolution runs of these same simulations are in progress, and preliminary results indicate that the LFs match well with what we see in our lower-resolution runs.

Interestingly, the host galaxies in our simulations typically have higher stellar masses than the observed galaxies for which their LF best matches. However, the virial masses are within the range estimated for MW-analog galaxies. It is common for simulations to produce an abundance of stars in the bulge of massive galaxies, but this discrepancy is unique to our massive hosts—the stellar mass-halo mass relation for dwarfs shows good agreement with observations, and will be published in forthcoming work. Since the virial masses are consistent with the Milky Way, and we deal only with the star-formation properties of dwarfs, we consider these appropriate MW-analogs for the purposes of this work.

To further ensure that our star-formation model accurately reproduces observed quenching in dwarf galaxies, we also measure mass trends in quenching of dwarf satellites. Here, and throughout this paper, we define a "quenched" halo as one with a specific star formation rate $\mathrm{SFR}/M_* < 10^{-11}$ yr^{-1}, where we define the SFR as the average rate of star-formation over the past 100 Myr. While many halos drop below this threshold only temporarily, most that do drop to $\mathrm{SFR} = 0$ $\mathrm{M}_\odot\,\mathrm{yr}^{-1}$,

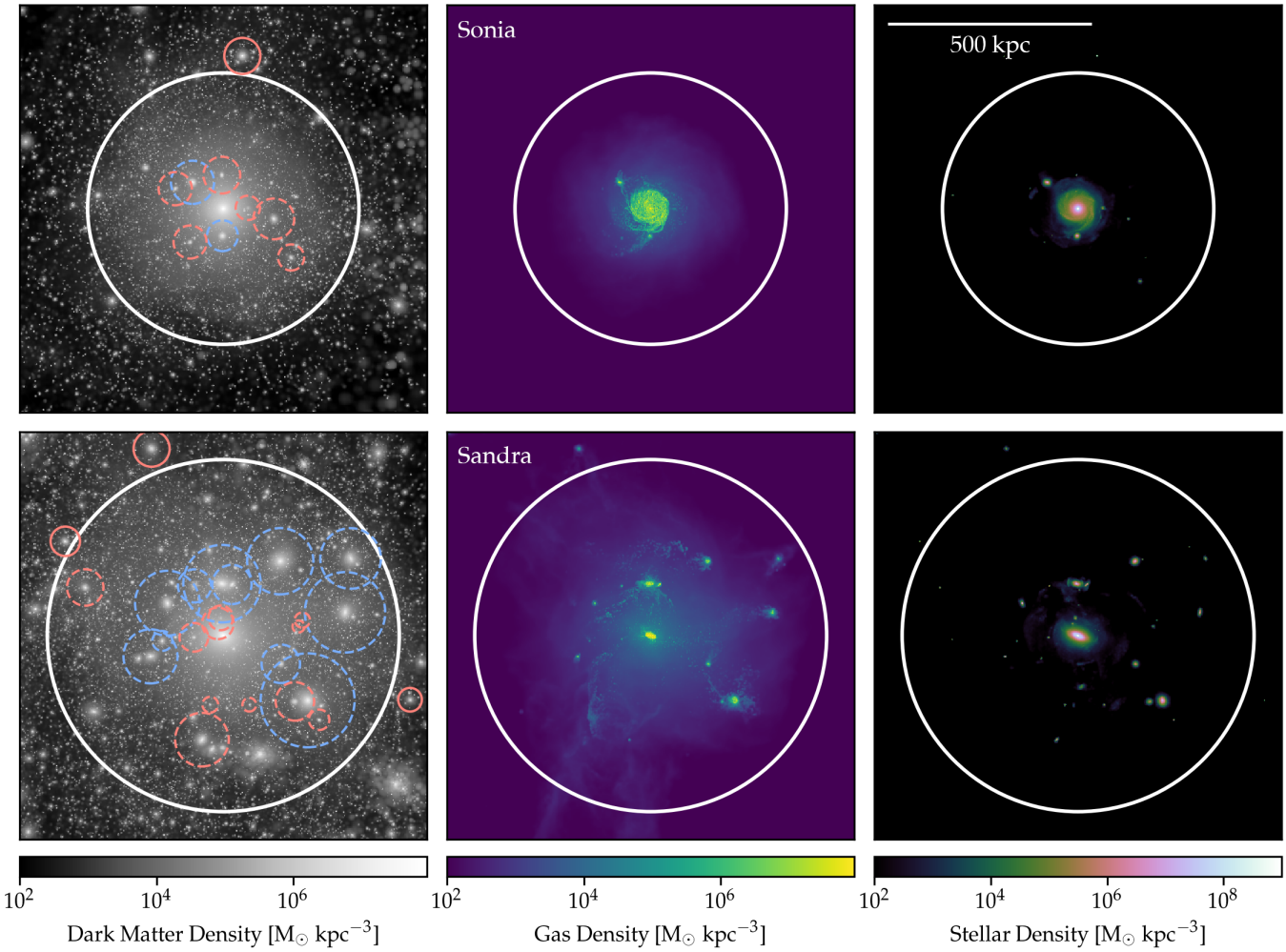

Figure 1. Line-of-sight averaged density plots of dark matter (left), gas (middle), and stars (right) in Sonia (top) and Sandra (bottom). Dark matter halos are outlined in the left panel, and the central halo is outlined for both gas and stars. Satellites are shown as dotted lines, and quenched galaxies are shown in red. Several dwarf satellites can be seen, many of which are gas-poor.

indicating genuine periods of quiescence. This choice of threshold has been adopted by many other studies using simulations (e.g. Wetzel et al. 2013; Bahé & McCarthy 2015; Pallero et al. 2019); however, observational studies often define quenching based on the atomic hydrogen richness $M_{\rm H\,I}/M_*$, a more readily measurable property. To facilitate observational comparisons, we compute the best-fit power-law relationship between the sSFR and atomic hydrogen richness in our simulations, finding that our sSFR threshold corresponds to $M_{\rm H\,I}/M_* \approx 0.2$. Thus, as it is necessary to use H I content for observational comparison, we will adopt this threshold.

Figure 3 shows the fraction of quenched satellites in each of the Justice League simulations, in 1 dex bins of stellar mass. In order to avoid sample contamination by galaxies far from the central host, we restrict our study to satellites within 300 kpc of their host halo. This is the typical estimate of the virial radii of the MW and M31, and corresponds to about 70–100% of the virial radii of our simulated hosts. Actual stellar masses are reported on this plot, which closely follow a mass-to-light ratio of 1 in the r-band.

Observational data from McConnachie (2012), Karachentsev et al. (2013), Weisz et al. (2015), and Geha et al. (2017) are shown for comparison. For the McConnachie (2012) data, we define the quenched fraction as the fraction of halos within 300 kpc of their host and with $M_{\rm H\,I}/M_* < 0.2$. Note that this threshold produces identical quenched fractions as the threshold of 0.1 adopted by Fillingham et al. (2015) and Wetzel et al. (2015), as the bimodal distribution of H I content means that the results are not sensitive to the exact choice of threshold. Similarly, Weisz et al. (2015) define a galaxy as a satellite if it is within 300 kpc of its Local Group host, but instead use the quenching time to define whether a galaxy is quenched. Weisz et al. (2015) also compile

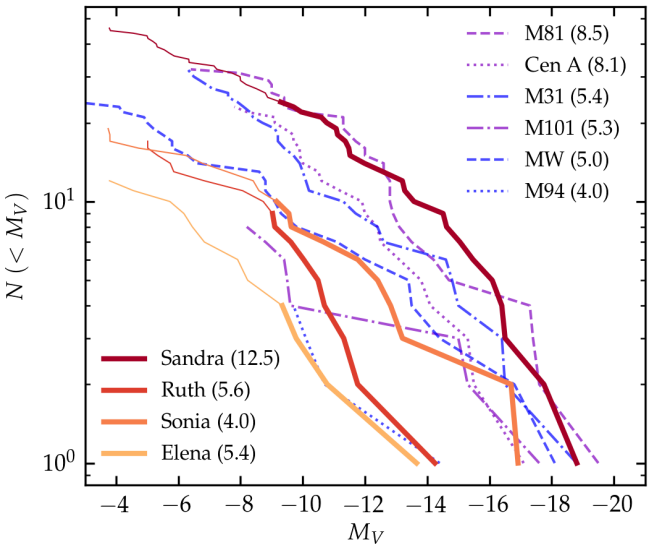

Figure 2. Cumulative satellite luminosity functions for the Justice League simulations. Red/orange lines show simulated LFs, with thin lines extending below the high-resolution limit of 100 star particles. Observational luminosity functions, presented for comparison, are shown with blue/purple and in dashed lines. Observational data comes from McConnachie (2012) for the Milky Way and M31, Crnojević et al. (2019) for Centaurus A, and Bennet et al. (2019) for M101. For M94, we include the two satellites recently discovered by Smercina et al. (2018), along with two previously known satellites KK 160 and IC 3687. Data for M81 comes from Chiboucas et al. (2013) and includes the recently discovered satellite from Smercina et al. (2017), using the $M_{r'}$ to M_V conversion provided by Crnojević et al. (2019). Values in parentheses are the host galaxy's stellar mass, in 10^{10} $M_\odot$ (for simulations, this is determined from M_R using a mass-to-light ratio of 1).

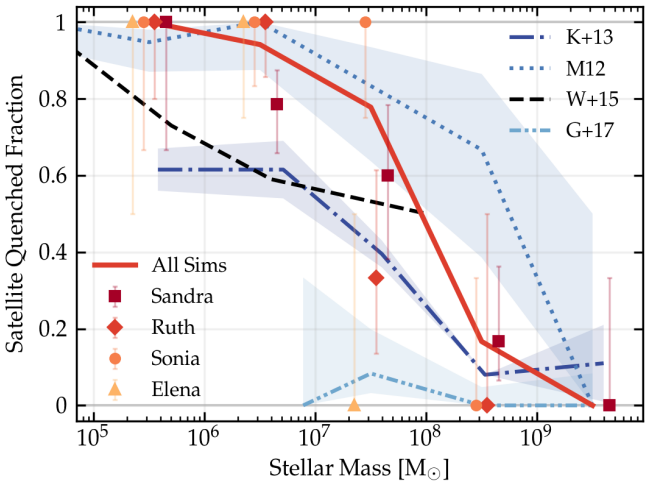

Figure 3. Quenched fractions for Justice League satellites in 1 dex bins of stellar mass. Individual simulations are indicated with points of varying shape, while the solid line represents the total for all four simulations. All simulated data are limited to galaxies with more than 100 star particles, and error bars represent 68% uncertainty in the binomial proportion via the Wilson Score Interval. Observational data from Geha et al. (2017, G+17), McConnachie (2012, M12), Weisz et al. (2015, W+15), and Karachentsev et al. (2013, K+13, as compiled by Weisz) are presented for comparison. Shaded regions represent the given uncertainty in observed quenched fractions. Our data generally agree with observations of the Local Group and other nearby galaxies, and a quenching–mass relationship is apparent.

data from the Nearby Galaxy catalog (Karachentsev et al. 2013), defining a galaxy as quenched based on its morphological classification. The upper and lower limits on their quenched fractions indicate varying whether dTrans galaxies are counted as quenched. Finally, Geha et al. (2017) use the presence of Hα in the galaxy spectrum to define quenching.

Figure 3 shows generally good agreement between quenched fractions in the Justice League simulations and nearby galaxy observations. Our simulations fall within the range of observations, fitting the data from McConnachie (2012) well at low masses and Weisz et al. (2015) and Karachentsev et al. (2013) better at higher masses. Given the widely varying methods used to define a galaxy as quenched, it is startling that our results agree as well as they do with Local Group observations, and suggest that morphology and H I content are well-correlated with star-formation rates. However, the SAGA survey (Geha et al. 2017) finds the majority of satellites to be star-forming, even at $M_* \sim 10^6$ $M_\odot$. While the SAGA survey includes hosts 20–40 Mpc away, McConnachie (2012) and Weisz et al. (2015) explore only the Local Group and Karachentsev et al. (2013) survey galaxies less than 11 Mpc away. These SAGA results are significantly different than ours and may suggest that the Local Group is an outlier with regards to dwarf satellite quenching.

Figure 3 also highlights variation between the individual Justice League simulations. The quenched fractions of the individual simulations vary widely, particularly at stellar masses of $\sim 10^{7-8}$ $M_\odot$. This halo-to-halo scatter is consistent with observed variations across the MW and M31, where we see drastic differences in quenched fractions. At $M_* \sim 10^{8-9}$ $M_\odot$, all M31 satellites (M32 & NGC 205) are quenched, while all MW satellites (the Magellenic clouds) are star-forming (McConnachie 2012).

It is apparent that our simulations accurately reproduce observed luminosity functions and quenched fractions of the Local Group and other nearby galaxies. We find a strong mass-dependence for the quenching of dwarf satellites, suggesting that environmental quenching may be more or less efficient at different mass scales.

In order to more directly assess quenching efficiency, we now investigate the quenching timescales of satellite galaxies in the Justice League simulations.

4. QUENCHING TIMESCALES

Having validated the Justice League simulations and established the mass-dependence of environmental quenching, we now endeavor to determine the relative times of quenching and infall. The quenching timescale, commonly defined as the time a satellite remains star-forming after infall, effectively measures the efficiency of environmental quenching and may hint at the underlying quenching processes.

4.1. *Timescale Definitions*

We define the quenching time ($t_{\rm quench}$) as the lookback time at which a galaxy last crossed below sSFR $= 10^{-11}$ yr^{-1}. While many studies (e.g. Weisz et al. 2015; Simpson et al. 2018; Fillingham et al. 2019) define $t_{\rm quench}$ as the time at which the galaxy has formed more than 90% of its present-day stellar mass, this choice is observationally motivated, as the sSFR is difficult to measure. As a check, we also compute $t_{\rm quench}$ as the time at which a satellite forms 99% of its present-day stellar mass. The resulting values are qualitatively unchanged from our sSFR definition, shifted to earlier times by a mean value of 0.34 Gyr.

To determine $t_{\rm quench}$, we compute each halo's SFH from its $z = 0$ stellar content. We use the $z = 0$ stellar content intentionally, as opposed to the instantaneous SFR of the main progenitor, to facilitate more direct comparisons to observations. We then divide the SFH by a spline fit to the stellar mass as a function of time, yielding the sSFR. Uncertainties in $t_{\rm quench}$ are computed from the quenching times with different choices of sSFR threshold, namely 0 yr^{-1} and 2×10^{-11} yr^{-1}. This is intended to flag those galaxies with particularly "strung-out" star formation histories, for which the exact quenching epoch is difficult to constrain.

We define the infall time ($t_{\rm infall}$) as the lookback time to the subhalo's first crossing of its host's virial radius ($R_{\rm vir}$). While many halos fall within 1 $R_{\rm vir}$ multiple times, we chose the first crossing because the traditional picture of stripping-dominated quenching attributes the majority of gas removal to the first pericentric passage (Slater & Bell 2013). As with $t_{\rm quench}$, uncertainties are computed by varying the choice of threshold, namely from 0.9 to 1.1 $R_{\rm vir}$.

4.2. *Results*

Figure 4 shows $t_{\rm quench}$ vs $t_{\rm infall}$ for all quenched halos surviving to $z = 0$ in the Justice League simulations.

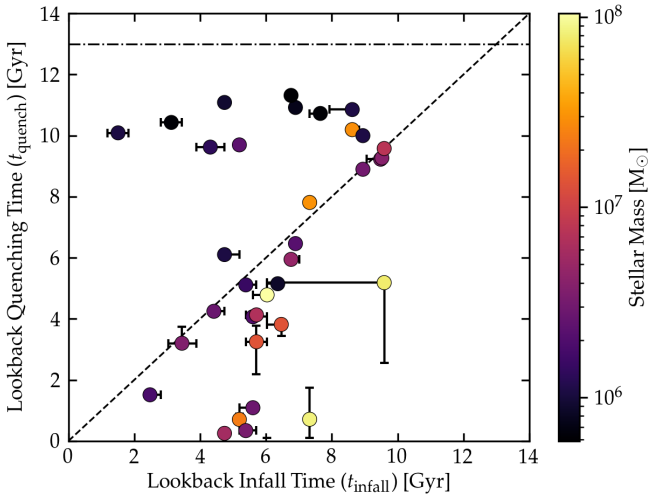

Figure 4. Lookback quenching time ($t_{\rm quench}$) vs lookback infall time ($t_{\rm infall}$) for all quenched halos surviving to $z = 0$ in our MW-analog simulations. Error bars indicate sensitivity to choice of threshold; that is, upper (lower) limits for quenching time use 2×10^{-11} yr^{-1} (0 yr^{-1}) rather than 10^{-11} yr^{1}, and upper (lower) limits for infall time use 1.1 (0.9) $R_{\rm vir}$ rather than 1 $R_{\rm vir}$. Points are colored by stellar mass, and halos with less than 100 star particles are excluded. The dashed line divides the plot into quenching before infall (left) and after infall (right), and the dot-dash line indicates the epoch of reionization. Many satellites quench after infall, and those that quench before infall do so largely independent of their infall time.

Galaxies falling to the right of the diagonal line on this plot were quenched after infall, while those to the left were quenched prior to infall. We see that several of the lowest-mass halos quench early on ($t_{\rm quench} \gtrsim 9$ Gyr), but with little correlation to infall time, indicating the presence of a non-environmental process. Importantly, we also see a population of predominantly higher-mass halos that quench after infall into the host halo (along and to the right of the diagonal line), to be discussed later.

Figure 4 also shows two satellites that quench shortly before infall into their host halo (just to the left of the diagonal line and with $t_{\rm quench} \lesssim 8$ Gyr). One is a reionization fossil that experienced momentary reignition prior to infall, but both seem to feel the effects of environment beyond 1 $R_{\rm vir}$. This extended environmental quenching is likely caused by some combination of unique orbital trajectory (e.g. infall along a gas filament), ram pressure stripping (even beyond the virial radius), and gas depletion (Bahé et al. 2013).

While some halos certainly do quench prior to infall, the higher-mass halos typically quench shortly after. This mass dependence is more readily apparent in Figure 5, which shows the quenching timescale

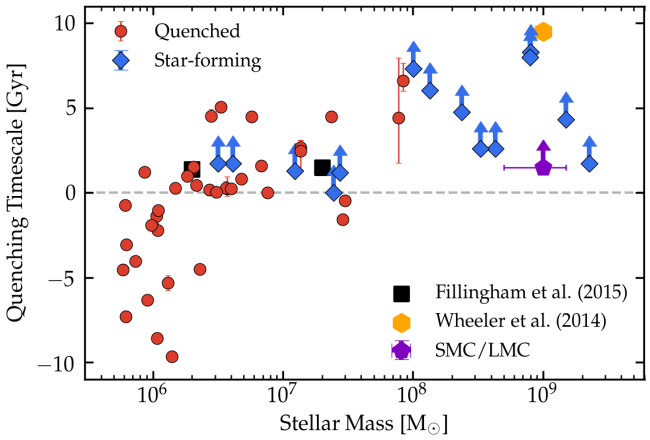

Figure 5. Quenching timescale ($t_{\rm infall} - t_{\rm quench}$) vs stellar mass. For halos that are quenched at $z = 0$ (red circles), this represents the time after infall until quenching, i.e. negative values indicate quenching prior to infall. For halos that are star-forming at $z = 0$ (blue diamonds), this can be interpreted physically as a lower limit on the possible quenching timescale. As in Figure 4, error bars indicate threshold variation (added in quadrature) and halos with less than 100 star particles are excluded. Observational estimates from Fillingham et al. (2015) and Wheeler et al. (2014) are shown in black squares and orange hexagons, respectively. An estimate of the quenching timescale for the SMC/LMC system is shown in purple, using the infall time estimate from Kallivayalil et al. (2013). A clear stellar mass threshold of $\sim 10^8$ M$_\odot$ is apparent, above which our satellites are largely resistant to rapid environmental quenching.

$t_{\rm infall} - t_{\rm quench}$ vs stellar mass. Galaxies which have become satellites but remain star-forming at $z = 0$ are shown on this plot, as their "quenching" timescale can be interpreted as a lower limit on the potential timescale. For intermediate-mass satellites ($M_* \sim 10^{6-7}$ M$_\odot$), we predominantly see rapid quenching timescales ($\lesssim 2$ Gyr) consistent with observational estimates of LG timescales (Wetzel et al. 2015; Fillingham et al. 2019). At higher masses ($M_* \gtrsim 10^8$ M$_\odot$), none of our satellites are quenched at $z = 0$, consistent with the unquenched Magellanic clouds, which experienced infall ~ 1.5 Gyr ago (Kallivayalil et al. 2013, their Figure 11). Additionally, our lower limits are largely below the quenching timescales estimated for this mass range by Wheeler et al. (2014).

Finally, a clear stellar mass threshold of 10^8 M$_\odot$ is apparent in Figure 5, above which satellites can remain star-forming for as much as ~ 7 Gyr after infall. Below this threshold, the only galaxies that remain star-forming are those which have only recently crossed $R_{\rm vir}$ ($\lesssim 1$ Gyr ago), and may still quench within the expected timescale. This threshold may indicate a characteristic mass scale at which the efficiency of the quenching process changes.

5. DISCUSSION

The quenching timescales we estimate for the Justice League satellites are generally consistent with observations of the Local Group. Timescales of $\lesssim 3$ Gyr for intermediate-mass satellites ($M_* \sim 10^{6-8}$ M$_\odot$) imply the role of an efficient environmental quenching process at this mass range. A likely candidate for this process is ram pressure stripping, which has been posited to rapidly remove gas from infalling satellites (Slater & Bell 2014; Wheeler et al. 2014; Wetzel et al. 2015), particularly with the aid of supernovae feedback processes (Caproni et al. 2015; Bahé & McCarthy 2015; Kazantzidis et al. 2017). The stellar mass threshold of $\sim 10^8$ M$_\odot$ is highly consistent with those determined observationally (e.g. Fillingham et al. 2015; Phillips et al. 2015), and implies that environmental quenching of satellites is only highly efficient below this mass. The inefficiency of quenching at these higher masses may indicate that strangulation is the dominant quenching mechanism (Wheeler et al. 2014; Wetzel et al. 2015). However, the processes governing quenching in dwarf satellites are likely highly interconnected. As such, definitive conclusions regarding the relative roles of quenching processes in the Justice League simulations will require more detailed analysis of gas inflow/outflow and host-satellite interactions.

Notably, our results are in conflict with the recent observations by the Satellites Around Galactic Analogs (SAGA) survey (Geha et al. 2017). Using Sunrise, we estimate that our stellar mass threshold of 10^8 M$_\odot$ corresponds to $M_r \sim -15.2$. Of the 27 satellites observed by SAGA so far, 26 are actively star-forming, despite many ($\sim 30\%$) having $M_r < -15.2$. It is worth noting that the SAGA completeness limit lies around $M_* \sim 10^6$ M$_\odot$ for star-forming galaxies and $\sim 10^7$ M$_\odot$ for quenched galaxies. Regardless, in the range $M_* \sim 7 \times 10^6$ –10^8 M$_\odot$, our simulations suggest a quenched fraction of 80%, not the 12.5% observed.

In the context of quenching timescales, several explanations may account for this discrepancy in quenched fractions. It may be that the SAGA sample so far is dominated by recently infalling satellites (i.e. the five star-forming satellites with $M_* < 10^8$ M$_\odot$ in Figure 5); however, this is unlikely to account for the entirety of the discrepancy, as many SAGA satellites are in close projected distances to their host. It may also be that quenching generally occurs more slowly, or less efficiently, in SAGA satellites, and that the Local Group is unique it its high quenching efficiency. However, the

nature of this discrepancy is difficult to constrain with small statistics—the complete SAGA survey of ~ 100 hosts will provide valuable data for understanding the efficiency of quenching outside of the Local Group.

Missing in this analysis is the quenched fractions and quenching timescales of ultra-faint dwarfs (UFDs; $M_* < 10^6$ $M_\odot$). Although the Justice League simulations are run at ultra-high resolution, these dwarfs approach our resolution limit and may be subject to additional uncertainty in the modeling of baryonic physics. Higher-resolution runs of the Justice League simulations are in progress, which, upon completion, will allow us to explore UFD quenching more thoroughly. Future papers (e.g. Applebaum et al. in prep) will explore quenching in ultra-faint dwarfs and the effects of reionization on these faint galaxies.

6. CONCLUSIONS

Using the ultra-high resolution Justice League simulations, we have explored the quenched fractions and quenching timescales of dwarf galaxies near MW-mass hosts. Our conclusions are summarized as follows:

1. The Justice League simulations accurately reproduce observations of the dwarf galaxies in the Local Group, both with regards to satellite luminosity functions and quenched fractions across 5 dex in stellar mass. While we see variability between individual simulations, this is largely consistent with observations.

2. We find that intermediate mass satellites ($M_* \sim 10^{6-7}$ $M_\odot$) generally have short quenching timescales of $\lesssim 3$ Gyr, indicating that quenching is very efficient at this mass.

3. We find a stellar mass threshold of $\sim 10^8$ $M_\odot$, above which infalling satellites are largely resistant to rapid environmental quenching and can remain star-forming for as much as 8 Gyr after infall. Below this threshold, only five satellites remain star-forming, all of which fell into their host halo ~ 1 Gyr. This may point to a characteristic mass scale at which the quenching process—and with it the quenching efficiency—shifts.

There is much further work to be done in analyzing these simulations. [A sentence that describes and cites relevant studies currently in-progress by collaborators.] Furthermore, the ultra-high resolution these simulations provide will facilitate detailed tracking of gas inflow and outflow, constraints on satellite contribution to the CGM, and analysis of observational signatures of quenching mechanisms. While it remains unclear the precise role that each quenching mechanism may play in the quenching of dwarf satellites, the Justice League simulations provide a promising route by which to better understand these processes.

The authors acknowledge Lucas Chamberland and Anna Engelhardt for fruitful discussions on the physics of dwarf satellite quenching. H.B.A. acknowledges support from the Grinnell College Mentored Advanced Project (MAP) program. This research is supported by NSF under CAREER grant No. AST-1848107.

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